

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 5: Factorisation

Topic-wise short notes with solved examples

Methods of factorising algebraic expressions: common factors, grouping, splitting the middle term, difference of two squares, perfect squares, and the sum and difference of two cubes.

## 1 What factorising means

- To **factorise** an expression is to write it as a product of simpler expressions (its factors). It is the reverse of expanding.
- Always look for a **common factor first**. Then choose a method from the number of terms.

| Number of terms | Try                                               |
|-----------------|---------------------------------------------------|
| Two             | Difference of squares; sum or difference of cubes |
| Three           | Perfect square; splitting the middle term         |
| Four or more    | Grouping; a perfect square with one extra term    |

- Factorise **completely**: keep going until no factor can be broken further.

## 2 Common factors

- Take out the HCF of all the terms. The common factor may itself be a bracket.

**Example 2.1** Factorise  $12a^2b - 18ab^2 + 6ab$ .

**SOLUTION**

$$6ab(2a - 3b + 1)$$

**Example 2.2** Factorise  $x(a - b) - y(b - a)$ .

**SOLUTION**

$$b - a = -(a - b), \text{ so the expression is } x(a - b) + y(a - b).$$

$$(a - b)(x + y)$$

## 3 Grouping

- Arrange the terms in groups so that each group has a common factor and the brackets left behind are the same.

**Example 3.1** Factorise  $ax + by + ay + bx$ .

SOLUTION

$$(ax + ay) + (bx + by) = a(x + y) + b(x + y)$$

$$(x + y)(a + b)$$

**Example 3.2** Factorise  $a^2 - ab - 3a + 3b$ .

SOLUTION

$$a(a - b) - 3(a - b)$$

$$(a - b)(a - 3)$$

**Example 3.3** Factorise  $x^2 + \frac{1}{x^2} + 2 - 5x - \frac{5}{x}$ .

SOLUTION

$$\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right)$$

$$\left(x + \frac{1}{x}\right)\left(x + \frac{1}{x} - 5\right)$$

## 4 Trinomials: splitting the middle term

- For  $x^2 + bx + c$ : find two numbers with **product**  $c$  and **sum**  $b$ .
- For  $ax^2 + bx + c$ : find two numbers with **product**  $ac$  and **sum**  $b$ , split the middle term, then group.
- If the product is positive the two numbers have the same sign as  $b$ . If it is negative they have opposite signs.
- If  $b^2 - 4ac$  is not a perfect square, the trinomial has no factors with rational coefficients.

**Example 4.1** Factorise  $x^2 + 7x + 12$ .

SOLUTION

Product 12, sum 7: the numbers are 3 and 4.

$$(x + 3)(x + 4)$$

**Example 4.2** Factorise  $x^2 - 5x - 24$ .

SOLUTION

Product  $-24$ , sum  $-5$ : the numbers are  $-8$  and  $3$ .

$$(x - 8)(x + 3)$$

**Example 4.3** Factorise  $6x^2 + 11x - 10$ .

**SOLUTION**

Product  $-60$ , sum  $11$ : the numbers are  $15$  and  $-4$ .

$$6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5)$$

$$(2x + 5)(3x - 2)$$

**Example 4.4** Factorise  $2(a + b)^2 - 5(a + b) - 3$ .

**SOLUTION**

Let  $y = a + b$ :  $2y^2 - 5y - 3 = (2y + 1)(y - 3)$

$$(2a + 2b + 1)(a + b - 3)$$

## 5 Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

- A **sum** of two squares  $a^2 + b^2$  has no real factors.
- Perfect-square trinomials:  $a^2 + 2ab + b^2 = (a + b)^2$  and  $a^2 - 2ab + b^2 = (a - b)^2$ .
- Four terms may form a perfect square and one more square: group three and one.

**Example 5.1** Factorise  $25x^2 - 49y^2$ .

**SOLUTION**

$$(5x + 7y)(5x - 7y)$$

**Example 5.2** Factorise  $a^4 - 16$ .

**SOLUTION**

$$(a^2 + 4)(a^2 - 4)$$

$$(a^2 + 4)(a + 2)(a - 2)$$

**Example 5.3** Factorise  $x^2 - 6x + 9 - 4y^2$ .

**SOLUTION**

$$(x - 3)^2 - (2y)^2$$

$$(x - 3 + 2y)(x - 3 - 2y)$$

**Example 5.4** Factorise  $x^4 + x^2 + 1$ .

**SOLUTION**

Add and subtract  $x^2$ :  $(x^2 + 1)^2 - x^2$

$$(x^2 + x + 1)(x^2 - x + 1)$$

**Example 5.5** Evaluate  $(6.7)^2 - (3.3)^2$ .

**SOLUTION**

$$(6.7 + 3.3)(6.7 - 3.3) = 10 \times 3.4$$

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## 6 Sum and difference of two cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2), \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

- The sign in the first bracket matches the sign in the expression. The middle sign in the second bracket is the opposite. The last sign is always plus.

**Example 6.1** Factorise  $8x^3 + 27y^3$ .

**SOLUTION**

$$(2x)^3 + (3y)^3$$

$$(2x + 3y)(4x^2 - 6xy + 9y^2)$$

**Example 6.2** Factorise  $a^3 - \frac{1}{a^3}$ .

**SOLUTION**

$$\left(a - \frac{1}{a}\right) \left(a^2 + 1 + \frac{1}{a^2}\right)$$

**Example 6.3** Factorise  $a^6 - b^6$ .

**SOLUTION**

$$(a^3 + b^3)(a^3 - b^3)$$

$$(a + b)(a - b)(a^2 - ab + b^2)(a^2 + ab + b^2)$$

**Example 6.4** Factorise  $2x^4 + 16x$ .

**SOLUTION**

$$2x(x^3 + 8)$$

$$2x(x + 2)(x^2 - 2x + 4)$$

## 7 Common mistakes

- Stopping early:  $a^4 - 16 = (a^2 + 4)(a^2 - 4)$  is not complete.
- Trying to factorise  $a^2 + b^2$ .
- Sign errors when a group starts with a minus:  $-3a + 3b = -3(a - b)$ .
- For  $ax^2 + bx + c$ , using numbers whose product is  $c$  instead of  $ac$ .
- Writing  $a^3 + b^3 = (a + b)(a^2 + ab + b^2)$ . The middle term is  $-ab$ .
- Not taking out the common factor first.

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